

RITSEM, Sweden

WMO: 020130

Lat: 67.7269N

Lon: 17.4711E

Elev: 1781

StdP: 13.77

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.8	(c) -8.1	(d) -17.4	(e) 2.3	(f) -11.4	(g) -13.5	(h) 2.8	(i) -7.6	(j) 27.1	(k) 23.3	(l) 23.9	(m) 21.3	(n) 2.3	(o) 30	(p) 0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	70.0	56.9	66.0	55.3	62.5	53.6	58.6	66.6	56.7	63.8	54.8	60.9	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.9	68.6	61.8	53.2	64.3	59.9	51.5	60.3	58.2	26.3	66.7	24.9	64.0	23.7	61.0	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 20.4	(b) 17.3	(c) 14.7	(e) -16.6	(f) 74.4	(g) 3.1	(h) 3.5	(i) -18.9	(j) 76.9	(k) -20.7	(l) 79.0	(m) -22.4	(n) 80.9	(o) -24.7	(p) 83.4
			(e) -17.1	(f) 61.3	(g) 3.2	(h) 2.0	(i) -19.4	(j) 62.7	(k) -21.2	(l) 63.9	(m) -23.0	(n) 65.0	(o) -25.3	(p) 66.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.4	14.5	14.6	19.1	27.9	37.9	47.1	53.7	51.6	43.6	33.0	24.6	20.2
	DBStd	15.96	11.46	11.36	9.72	7.37	6.87	6.48	5.97	5.76	5.77	6.90	8.20	10.35
	HDD50	6724	1100	992	958	662	383	133	29	48	203	528	763	925
	HDD65	11896	1565	1412	1423	1112	841	538	353	415	642	992	1213	1390
	CDD50	305	0	0	0	0	6	45	144	99	11	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	21	0	0	0	0	0	3	15	3	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	6.2	5.9	5.8	6.1	6.2	6.5	6.0	5.6	5.6	6.3	6.8	6.6	6.5
	PrecAvg	20.4	2.0	1.5	1.4	0.9	1.2	1.5	2.4	2.2	1.9	1.9	1.5	1.9
(15) Precipitation	PrecMax	24.6	6.8	3.5	3.1	1.8	2.4	2.2	5.5	4.1	4.0	4.1	3.7	5.0
	PrecMin	16.7	0.1	0.6	0.4	0.2	0.4	0.2	0.9	0.9	0.5	0.5	0.3	0.4
	PrecStd	2.3	1.3	0.8	0.8	0.5	0.5	0.6	1.3	0.8	1.0	0.8	1.0	1.1
	PrecStd	2.3	1.3	0.8	0.8	0.5	0.5	0.6	1.3	0.8	1.0	0.8	1.0	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	37.6	37.9	39.5	47.3	65.6	70.6	76.2	72.7	60.9	51.5	43.0	39.1	39.1
	0.4% MCWB	35.2	35.6	34.5	39.9	50.5	54.4	60.0	57.8	51.8	46.5	39.9	36.1	36.1
	2% DB	35.3	35.2	36.7	43.2	56.7	64.9	71.4	68.5	57.0	46.7	40.3	37.0	37.0
	2% MCWB	33.1	32.6	33.5	37.8	46.8	52.1	58.3	56.7	50.2	43.3	37.9	34.4	34.4
	5% DB	33.1	33.1	34.4	40.3	51.7	60.9	67.3	64.2	54.2	44.7	37.4	35.1	35.1
	5% MCWB	31.4	31.0	31.7	35.9	43.7	50.5	56.4	55.5	49.1	42.1	35.1	32.7	32.7
	10% DB	30.2	30.6	32.0	37.8	47.2	57.7	63.6	60.8	52.0	42.3	35.1	32.8	32.8
	10% MCWB	28.5	28.8	29.5	34.2	41.0	49.1	54.7	53.7	47.7	39.6	33.0	30.5	30.5
	0.4% WB	35.7	36.0	35.7	40.6	51.4	56.6	61.9	60.4	53.8	47.0	40.5	36.8	36.8
	0.4% MCDB	37.1	37.7	38.1	45.6	63.7	64.8	72.8	69.5	58.6	51.0	42.5	38.4	38.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.5	33.1	33.8	38.2	47.4	54.0	59.7	58.0	51.5	44.2	38.2	34.6	34.6
	2% MCDB	34.8	34.7	36.1	42.6	56.0	61.6	68.8	65.1	55.6	46.7	40.0	36.4	36.4
	5% WB	31.7	31.2	31.9	36.3	44.3	51.8	57.6	56.3	49.6	42.0	35.5	32.5	32.5
	5% MCDB	33.1	33.0	34.3	39.9	51.0	59.9	65.5	63.0	53.5	44.4	37.3	34.3	34.3
	10% WB	28.4	28.8	29.6	34.4	41.6	49.8	55.7	54.5	48.0	39.8	33.2	30.3	30.3
	10% MCDB	30.4	30.6	31.9	37.5	46.7	57.1	62.0	60.2	51.7	42.0	34.9	32.1	32.1
(35) Mean Daily Temperature Range	MDBR	11.2	10.9	11.2	10.7	10.3	11.6	12.3	11.3	9.3	7.6	8.4	9.7	9.7
	5% DB	10.9	10.7	10.3	10.8	17.1	17.7	18.4	16.8	12.5	8.7	8.7	9.8	9.8
	5% WB	9.7	10.0	8.4	7.5	9.6	9.4	9.0	8.3	7.9	6.8	7.2	8.5	8.5
	MCDBR	10.1	11.1	10.2	10.3	16.2	16.0	16.4	15.2	11.2	8.6	8.3	9.9	9.9
	MCWBR	9.4	10.5	8.8	7.4	9.4	9.4	9.0	8.2	7.8	6.9	7.1	8.9	8.9
	MCWBR	9.4	10.5	8.8	7.4	9.4	9.4	9.0	8.2	7.8	6.9	7.1	8.9	8.9
(40) Clear-Sky Solar Irradiance	taub	0.167	0.227	0.268	0.292	0.305	0.311	0.320	0.308	0.281	0.242	0.171	N/A	N/A
	taud	1.924	2.141	2.213	2.249	2.316	2.464	2.517	2.531	2.562	2.279	1.958	N/A	N/A
	Ebn at Noon	50	193	248	274	282	284	278	269	252	186	43	0	0
	Edh at Noon	3	14	24	31	33	30	27	24	18	12	3	0	0
	RadAvg	12	125	518	1136	1544	1556	1407	1041	545	182	31	0	0
(45) All-Sky Solar Radiation	RadStd	1	26	80	83	137	181	152	114	58	34	3	0	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page